

XRA-GS: X-ray Attenuation-Aligned Gaussian Splatting for Sparse Tomographic View Synthesis

Abstract

*3D Gaussian Splatting has been extended to X-ray novel view synthesis, yet existing methods retain the gradient-driven densification rules from natural-image 3DGS without modification. Under sparse CT acquisition, these rules cause Gaussians to accumulate near high-contrast anatomical boundaries while interior tissues that determine X-ray attenuation remain under-represented. We present **XRA-GS**, an X-ray Attenuation-Aligned Gaussian Splatting framework that corrects this structural imbalance by redesigning the Gaussian evolution pipeline. XRA-GS introduces three components: a support-profile seeding strategy that initializes Gaussians from the coarse attenuation support estimated by FDK reconstruction; a geometry-aware pruning step that removes Gaussians concentrated near anatomical boundaries; and an adaptive density modulation step that applies position-dependent density refinement throughout training. Experiments on five organs under a sparse-view CT protocol demonstrate that XRA-GS achieves state-of-the-art novel view synthesis quality against six representative baselines from both visible-light sparse-view synthesis and X-ray CT methods.*

1. Introduction

Computed tomography (CT) is an essential imaging technique for noninvasively examining the internal structure of an object, and is widely used in clinical diagnosis, image-guided intervention, and industrial inspection [FDK84; AK84; WYD08]. During a CT acquisition, an X-ray source rotates around the object while a detector records multi-angle attenuation measurements, from which cross-sectional images or projections at unseen angles can be re-

covered through tomographic reconstruction. Reducing the number of acquired views is highly desirable, since it directly lowers radiation dose, shortens scan time, and broadens applicability to patients who cannot sustain long, stable acquisitions [LMC*23; WHL*24]. In this work, we study *Sparse Tomographic View Synthesis*: from only a handful of X-ray projections, the model must predict the line-integrated attenuation observed at unseen acquisition angles, so that each synthesized projection remains consistent with the Beer–Lambert imaging ge-

ometry rather than merely visually similar to the inputs. A reliable solution supplies auxiliary views for anatomical localization, lesion-range inspection, and intraoperative planning under severely limited sampling, where conventional analytical reconstruction breaks down [WN19].

Visible-light and X-ray imaging are governed by fundamentally different rendering equations, and this difference dictates where representation capacity should reside. In visible-light 3DGS, a pixel is formed by front-to-back α -compositing of an emission-absorption volume rendering integral [KKLD23; Max95]: the accumulated transmittance decays multiplicatively along the ray, so for opaque scenes the integrand is dominated by Gaussians near the first-hit surface while contributions from deeper Gaussians are progressively masked. The surface-clustered Gaussian layout that emerges under α -compositing is therefore a direct consequence of this transmittance-induced concentration, and is consistent with the surface-aligned geometry that surface-oriented 3DGS variants exploit explicitly [GL24; HYC*24]. In X-ray imaging, however, the detector reading follows the Beer–Lambert law [KS01]: after taking the logarithm, the recorded quantity becomes a linear, order-independent line integral of the attenuation coefficient along the ray, with no transmittance decay and no occlusion. Consequently, every Gaussian on the path contributes on equal footing, and capacity must be distributed *along the entire ray path* rather than collapsed onto any single layer. Existing X-ray Gaussian methods such as X-Gaussian, R2-Gaussian, X-Field, and DGR have correctly replaced α -compositing with a radiative line-integral renderer [CLW*24; ZLC*24; WTW*25; WLW*25], and sparse-view 3DGS pipelines such as FSGS, CoR-GS, and DNGaussian further compensate for missing observations through depth or co-

regularization priors [ZFJW24; ZLY*24; LZB*24]. Yet all of them inherit a Gaussian evolution rule that was originally designed for surface-centric α -compositing [KKLD23; RNSZ24].

Crucially, the issue is not gradient-driven densification itself. Splitting or cloning Gaussians where the projection-error gradient is large is a sound idea, and our method retains exactly this mechanism. The issue is that under a line-integral renderer, the residual gradient is still concentrated at high-attenuation-contrast interfaces, such as bone–soft-tissue boundaries, which almost coincide with the high-color-contrast interfaces that drive visible-light densification. A naïve port therefore keeps spawning Gaussians at anatomical boundaries even though the rendering equation now asks for capacity *along* the path. Under sparse views, this bias surfaces not as insufficient coverage but as a structural *capacity misallocation*: a substantial fraction of Gaussians is spent on boundary-driven redundancy, while interior path segments that determine the attenuation integral remain under-represented. The top row of Fig. 1 shows the natural fit between α -compositing and surface-clustered Gaussians; the bottom row shows that when the same evolution rule is reused under line-integral rendering, capacity still collapses to boundaries and leaves the deep path empty. Fixing this requires *augmenting*, not discarding, gradient-driven densification with attenuation-aligned decisions about where to seed, what to prune, and how to refine.

Motivated by this analysis, we propose **XRA-GS**, an X-ray Attenuation-Aligned Gaussian Splatting framework for Sparse Tomographic View Synthesis. Rather than appending auxiliary branches to a conventional densification backbone, **XRA-GS** keeps the gradient-driven backbone of standard 3DGS and adds three attenuation-aligned stages that

act on *where* Gaussians are seeded, retained, and refined, as illustrated in Fig. 1. Support-Profile Seeding (SPS) anchors initialization to a coarse FDK attenuation support [FDK84], so that Gaussians populate physically meaningful X-ray paths instead of starting from a contrast-biased prior; Geometry-aware Pruning (GAP) recycles Gaussians that accumulate near high-contrast anatomical interfaces during training, draining boundary-driven redundancy at its source; and Adaptive Density Modulation (ADM) applies position-dependent online refinement so that retained Gaussians carry interior attenuation rather than uniformly inflated density. Across diverse organs and view budgets, this attenuation-aligned evolution reallocates representation capacity from boundary redundancy to attenuation-relevant interior structures.

Our main contributions are summarized as follows:

- We identify *capacity misallocation* as the failure mode when sparse-view X-ray Gaussian methods inherit a visible-light evolution rule under a line-integral renderer: gradient-driven densification keeps clustering Gaussians at attenuation boundaries while interior path segments that determine the line integral remain starved of capacity.
- We propose **XRA-GS**, an X-ray Attenuation-Aligned Gaussian Splatting framework that augments standard gradient-driven densification with three coordinated attenuation-aligned stages—Support-Profile Seeding (SPS) for path-anchored initialization from coarse FDK support, Geometry-aware Pruning (GAP) for boundary-redundancy recovery, and Adaptive Density Modulation (ADM) for position-dependent online refinement—without modifying the X-ray renderer.

- Extensive experiments on multi-organ sparse-view CT show that **XRA-GS** delivers consistent gains in projection fidelity over state-of-the-art X-ray Gaussian and sparse-view 3DGS base-lines, while reducing the number of Gaussians and training cost; ablations confirm that SPS, GAP, and ADM each contribute complementary capacity reallocation.

2. Related Work

2.1. Gaussian Splatting

3D Gaussian Splatting (3DGS) represents a scene with a set of anisotropic Gaussians and renders novel views in real time through differentiable tile-based rasterization [KKLD23]. Building on this explicit representation, subsequent work has expanded along several directions: surface-aligned variants that recover geometrically accurate meshes from Gaussians [HYC*24; GL24]; dynamic and deformable extensions for time-varying scenes [WRW*24; YGZ*24]; dense visual SLAM systems built on top of the Gaussian representation [YQX*24]; generative pipelines that lift 3DGS into 3D content creation [TRZ*24]; and compactness/efficiency redesigns that reduce the Gaussian count or stabilize rendering across scales [NMKW24; LYX*24; YCH*24]. Across all these directions, the underlying evolution rule remains the gradient-driven clone-and-split densification together with spherical-harmonics view-dependent color, both of which are tailored to dense-view, surface-reflectance scenes. When either assumption is violated—inputs become sparse, or the imaging model switches from surface radiance to line-integral attenuation—this rule starts to misallocate Gaussian capacity, and the next two subsections examine these two regimes in turn.

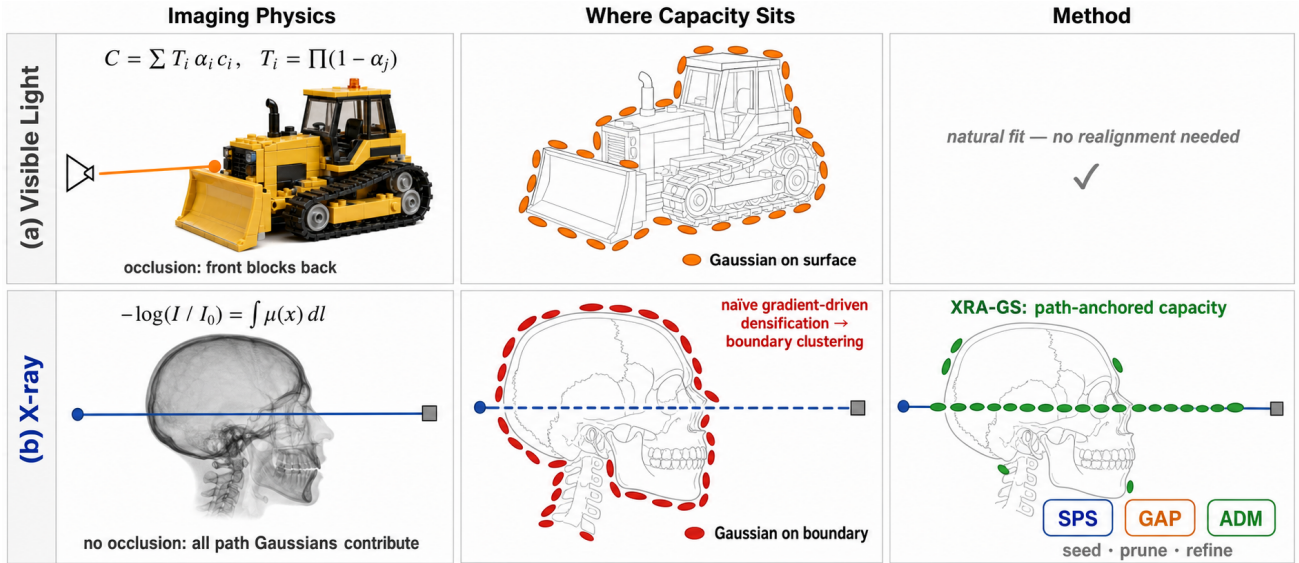


Figure 1: Visible-light vs. X-ray Gaussian rendering and the capacity gap they impose. (a) Visible-light α -compositing concentrates capacity on visible surfaces, where front Gaussians occlude back ones. (b) X-ray line integrals weight every Gaussian along the ray equally, so naïve gradient-driven densification still clusters at attenuation boundaries; **XRA-GS** realigns the evolution rule through SPS, GAP, and ADM to anchor capacity along the path.

2.2. Sparse View Synthesis

Neural Radiance Fields (NeRF) recover a continuous volumetric field through coordinate MLPs and volume rendering [MST*21], and degrade noticeably when only a handful of input views are available. NeRF-side sparse-view methods approach this limitation from complementary angles, each adding a different external prior that has been empirically validated: depth supervision from sparse structure-from-motion points [DLZR22; WCLL23], semantic consistency from pretrained vision encoders [JTA21], and geometric/appearance regularization with frequency-domain annealing [NBM*22; YPW23]. Sparse-view 3DGS work explores a similarly diverse set of perspectives on the explicit Gaussian set: FSGS attributes degradation to insufficient coverage and augments densification with proximity-guided unpooling and depth pri-

ors [ZFJW24]; CoR-GS treats it as unstable optimization and adds dual-field co-regularization with co-pruning [ZLY*24]; DNGaussian injects geometry through depth normalization [LZB*24]; and DropGaussian counters overfitting via stochastic Gaussian dropout [PRK25]. A complementary feed-forward line further moves geometry completion into large-scale pretraining [CLTS24; CXZ*24; XPW*25; LFY*25]. These different angles all yield empirical gains, which suggests that sparse-view degradation admits multiple compatible explanations rather than a single root cause. The next subsection turns to how that assumption itself is challenged under X-ray line integration.

2.3. Tomographic Novel View Synthesis

Sparse-view tomographic novel view synthesis aims to recover unseen X-ray projections, and the un-

derlying density volume, from a small number of input projections [WYD08; WN19], in contrast to analytical (FDK) and iterative (SART) reconstructions that either suffer severe streak artifacts under sparse views or require many iterations [FDK84; AK84]. Learning-based methods fall into two families with complementary entry points. Implicit methods adapt neural attenuation fields to X-ray imaging—through self-supervised sinogram synthesis, hash-encoded acceleration, transformer-based structural modeling, or population-level encoder–decoder inference [ZIL*21; GWS*24; CWY*24; LFL*25]. Explicit methods redesign Gaussians around X-ray attenuation, covering radiative kernels with X-ray rasterization, integration-bias correction with differentiable voxelization, and material-aware ellipsoids with explicit segment-length physics, alongside further extensions toward discretized voxels, artifact-suppression graphs, 4D dynamics, and slice-wise reconstruction [CLW*24; ZLC*24; WTW*25; WLG*25; YMS*25; YCZ*25; Kua*25] (the last four target imaging regimes outside our sparse-view novel-view protocol and are omitted from the main comparison). Across both families, improvements have concentrated on the imaging equation or the kernel itself rather than on the evolution rule that decides where Gaussians accumulate, leaving sparse-view CT still bottlenecked by capacity misallocation inherited from visible-light densification. **XRA-GS** addresses this evolution rule directly, with SPS, GAP, and ADM aligning initialization, pruning, and refinement to the X-ray attenuation path.

3. Method

3.1. Preliminary

3.1.0.1. X-ray attenuation imaging. An X-ray detector pixel records the residual energy of a single ray after traversing the object, governed by the Beer–

Lambert law. For a ray $\mathbf{r}$ with source intensity I_0 and attenuation field $\mu(\mathbf{x})$, the standard practice is to work in the logarithmic domain [KS01], so that each pixel becomes a line integral of attenuation along the ray:

$$I(\mathbf{r}) = -\log \frac{I'(\mathbf{r})}{I_0} = \int_{\mathbf{r}} \mu(\mathbf{x}) d\mathbf{l}. \quad (1)$$

Unlike α -compositing in visible-light rendering, this integral is linear and order-independent: every point along $\mathbf{r}$ contributes additively, with no occlusion between front and back materials.

3.1.0.2. Radiative Gaussian backbone. We represent the scanned object with a set of radiative Gaussians $\mathcal{G} = \{G_i\}_{i=1}^N$, following the recent line of explicit Gaussian-based tomographic representations [CLW*24; ZLC*24]. Each Gaussian G_i is parameterized by a center $\mathbf{p}_i$, a covariance Σ_i , and a non-negative density ρ_i , defining a local density field

$$G_i(\mathbf{x}) = \rho_i \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{p}_i)^\top \Sigma_i^{-1}(\mathbf{x} - \mathbf{p}_i)\right), \quad (2)$$

and the global attenuation field is the sum $\mu(\mathbf{x}) = \sum_{i=1}^N G_i(\mathbf{x})$. Substituting this representation into Eq. (1) admits a differentiable X-ray rasterizer $\mathcal{R}$, so that for any acquisition geometry π_v a projection can be rendered as

$$\hat{I}_v = \mathcal{R}(\mathcal{G}; \pi_v). \quad (3)$$

3.1.0.3. Problem statement. Given a sparse-view observation set $\mathcal{P} = \{(I_v, \pi_v)\}_{v=1}^V$ and a coarse FDK reconstruction V_{FDK} , our goal is to learn $\mathcal{G}$ in a per-case optimization setting so that target projections at unseen angles can be synthesized from only a few observed projections. The radiative backbone has already aligned the renderer with X-ray physics; what remains misaligned in this regime is the evolution rule that shapes $\mathcal{G}$ during training, where gradient-driven densification continues to concentrate Gaussians at high-gradient anatomical

boundaries. **XRA-GS** realigns this evolution rule through three modules introduced below, while leaving the rasterizer and the differentiable voxelizer of R2-Gaussian [ZLC*24] unchanged.

3.2. Overall Framework

As illustrated in Figure 2, **XRA-GS** instantiates the attenuation-aligned evolution pipeline as three modules acting at three distinct stages of optimization: SPS replaces standard initialization by seeding Gaussians along the FDK attenuation support; GAP acts inside the densification loop and reclaims locally redundant Gaussians produced by gradient-driven growth; ADM acts on retained Gaussians and refines their density values using position-dependent spatial context. At inference, the rasterizer remains identical to the radiative backbone, so the added cost is concentrated in training.

3.3. Support-Profile Seeding (SPS)

Initialization strongly conditions the optimization trajectory of radiative Gaussians, yet standard 3DGS initializers transfer poorly to X-ray tomography: uniform sampling ignores the strong contrast between air, soft tissue, and bone, while SfM point-cloud initialization is unstable under gray-value overlap and weak local texture in X-ray projection [CLW*24]. The sparse-view CT literature has consistently treated the coarse FDK volume V_{FDK} as carrying reliable foreground support and low-frequency attenuation layout, even when its fine detail is corrupted by streak artifacts [WCS*22; LMC*23; LFZ*25; ZLC*24]. We exploit this signal as a sampling prior rather than as a target volume. Concretely, over the voxel grid Ω underlying V_{FDK} we sample seed positions from a density-weighted mix-

ture distribution

$$q(\mathbf{x}) = \alpha \frac{1}{|\Omega|} + (1-\alpha) \frac{V_{\text{FDK}}(\mathbf{x})^\gamma}{Z}, \quad Z = \sum_{\mathbf{x}' \in \Omega} V_{\text{FDK}}(\mathbf{x}')^\gamma, \quad (4)$$

that linearly interpolates between a uniform prior and a power-shaped attenuation profile. Per-Gaussian densities are then queried from V_{FDK} at the sampled locations and per-Gaussian scales are set by nearest-neighbor spacing. The mixture form is critical: density weighting alone would collapse capacity onto the brightest bony structures, while the uniform term retains global coverage so that under-attenuating but anatomically relevant regions remain populated.

3.4. Geometry-aware Pruning (GAP)

Inside the radiative backbone, projection-error gradients are systematically larger near bone–soft-tissue boundaries, so standard gradient-driven densification keeps duplicating Gaussians around the same high-contrast surfaces. Large gradients in this regime do not always imply missing coverage: they can also reflect that an already-saturated boundary is being represented redundantly by neighboring Gaussians. Rather than adding more support points in sparse regions in the spirit of FSGS [ZFJW24], GAP acts in the opposite direction and reclaims locally crowded Gaussians during densification.

For each Gaussian G_i we define a proximity score from its K nearest neighbors $\mathcal{N}_K(i)$,

$$s_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} \|\mathbf{p}_i - \mathbf{p}_j\|_2, \quad (5)$$

and combine it with $\bar{g}_i$, the accumulated positional-gradient magnitude of G_i over the current densification cycle. A Gaussian is flagged as a redundancy candidate only when it is simultaneously locally crowded and recently inactive:

$$m_i = \mathbf{1}[s_i \geq \tau \vee \bar{g}_i \geq \delta], \quad (6)$$

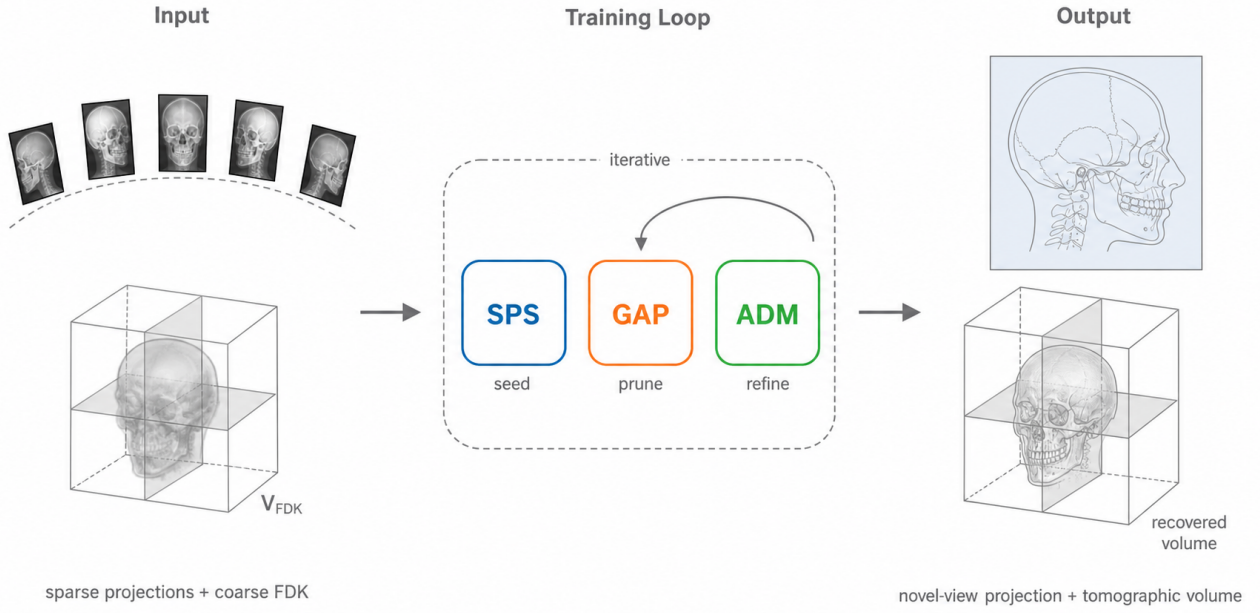


Figure 2: Overall framework of XRA-GS. Sparse X-ray projections together with a coarse FDK reconstruction enter a radiative Gaussian backbone whose Gaussian evolution is reorganized by SPS (seed), GAP (prune), and ADM (refine). The three modules act at initialization, inside the densification loop, and during later optimization, respectively, leaving inference unchanged.

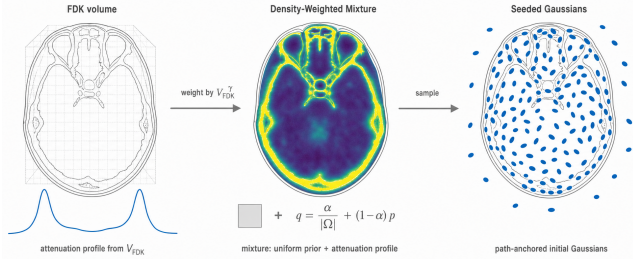


Figure 3: Support-Profile Seeding (SPS). From the coarse FDK volume, SPS builds a density-weighted mixture distribution that interpolates between a uniform prior and the attenuation profile. Sampling from this mixture seeds Gaussians along the anatomical support before any optimization step.

i.e. $m_i = 0$ marks a prune candidate. GAP then removes a bounded fraction of the flagged Gaussians

per cycle and slightly contracts the covariance of their retained neighbors to avoid creating local support holes. The dual criterion is what distinguishes GAP from purely density-based pruning: gradient-activity prevents deleting Gaussians that are still learning a boundary, while proximity prevents over-densified regions from continuously monopolizing capacity.

3.5. Adaptive Density Modulation (ADM)

Even after path-anchored initialization and redundancy control, density values across anatomy remain heterogeneous: bone exhibits higher attenuation, air-filled regions should stay near zero, and soft tissue requires smoother transitions. A single per-Gaussian gradient step has difficulty expressing this spatial heterogeneity without overshooting one

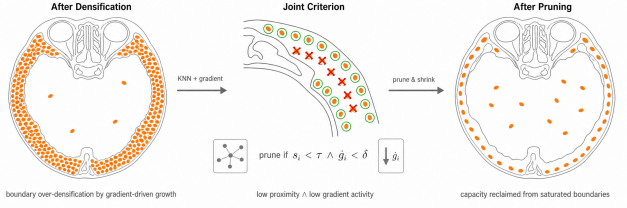


Figure 4: Geometry-aware Pruning (GAP).

After gradient-driven densification, Gaussians cluster around high-contrast boundaries. GAP flags a Gaussian as a redundancy candidate only when it is both locally crowded ($s_i < \tau$) and recently inactive ($\bar{g}_i < \delta$), so that capacity is reclaimed from saturated boundaries without removing Gaussians that are still learning structure.

regime while undershooting another. ADM therefore encodes position-dependent spatial context through K-Planes [FYT*23] and uses it to predict a relative density correction, without touching Gaussian centers or covariances.

Given a position $\mathbf{x}$, we bilinearly sample three orthogonal feature planes $\{P_{uv}\}_{uv \in \{xy, xz, yz\}}$ at the corresponding 2D projections $\Pi_{uv}(\mathbf{x})$ and concatenate the results,

$$F(\mathbf{x}) = \text{concat}(\mathcal{B}(P_{xy}, \Pi_{xy}(\mathbf{x})), \mathcal{B}(P_{xz}, \Pi_{xz}(\mathbf{x})), \mathcal{B}(P_{yz}, \Pi_{yz}(\mathbf{x}))), \quad (7)$$

where $\mathcal{B}(\cdot)$ denotes bilinear sampling. A lightweight dual-head MLP h decodes $F(\mathbf{x})$ into a density offset $\Delta\sigma(\mathbf{x})$ and a confidence gate $g(\mathbf{x}) \in [0, 1]$. We center the offset by its batch mean $\overline{\Delta\sigma}$ and modulate the base density multiplicatively:

$$\rho_{\text{final}}(\mathbf{x}) = \rho_{\text{base}}(\mathbf{x}) \cdot \left(1 + g(\mathbf{x}) \cdot (\Delta\sigma(\mathbf{x}) - \overline{\Delta\sigma})\right). \quad (8)$$

The plane features and the MLP parameters are trained end-to-end on the current case under the rendering loss alone, with no extra density supervision; the modulation amplitude is attenuated when fewer

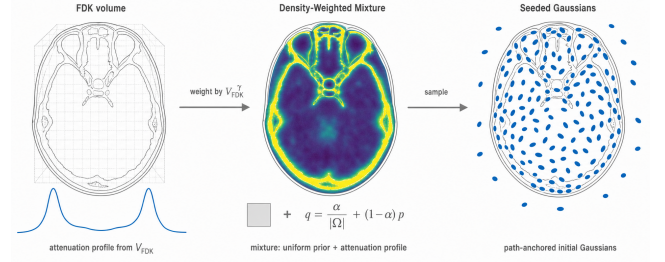


Figure 5: Adaptive Density Modulation (ADM).

For each Gaussian center, three orthogonal K-Planes feature planes are bilinearly sampled and concatenated; a dual-head MLP outputs a density offset and a confidence gate, which together apply a batch-centered relative correction $\rho_{\text{final}} = \rho_{\text{base}}(1 + g(\Delta\sigma - \overline{\Delta\sigma}))$.

input views are available and is ramped in only after the backbone has reached a stable geometric estimate. Batch centering turns $\Delta\sigma$ into a relative correction rather than a global density boost, so ADM amplifies density where the attenuation support warrants finer detail and damps modulation where the evidence is weak.

3.6. Training Loss

XRA-GS is supervised end-to-end on the observed views by a single objective combining projection-domain reconstruction with regularization on the volumetric density and the K-Planes features:

$$\mathcal{L} = \mathcal{L}_1 + \lambda_d \mathcal{L}_{\text{dssim}} + \lambda_v \mathcal{L}_{3\text{DTV}} + \lambda_p \mathcal{L}_{\text{pTV}}. \quad (9)$$

The first two terms are the standard L_1 and D -SSIM losses between rendered and observed projections; $\mathcal{L}_{3\text{DTV}}$ is a total-variation prior on the volumetric density retrieved by the differentiable voxelizer of R2-Gaussian, and $\mathcal{L}_{\text{pTV}}$ is a planar total-variation prior on the K-Planes features used by ADM. The K-Planes features and the MLP parameters of ADM

are jointly updated with the backbone Gaussian parameters under $\mathcal{L}$, so density modulation is learned online rather than fixed in advance.

4. Experiments

4.1. Experimental Setup

Data and task setting. We evaluate on the five-organ dataset from the X-Gaussian repository [CLW*24]: Chest (LIDC-IDRI) and Head, Abdomen, Foot, and Pancreas (open scientific visualization dataset). Following the NAF/X-Gaussian preprocessing and projection protocol, we construct sparse-view settings by retaining only 2, 3, or 4 input projections per scene. Models are trained exclusively on observed projections and evaluated on held-out novel angles.

Baselines. We compare against six methods in two groups. *Direct X-ray/CT baselines:* X-Gaussian [CLW*24] (radiative Gaussian splatting), R2-Gaussian [ZLC*24] (reconstruction-consistency regularization), and X-Field [WTW*25] (material-aware implicit modeling) represent the state of the art for X-ray novel view synthesis. *Natural-scene sparse-view 3DGS references:* CoR-GS [ZLY*24], DNGaussian [LZB*24], and FSGS [ZFJW24] characterize how gradient-driven densification degrades under tomographic projection. X-Field is re-evaluated under our identical protocol and denoted as X-Field* in the result tables and figures.

Metrics. Following the official X-Gaussian evaluation script, PSNR2D and SSIM2D are computed on $[0, 1]$ -normalized projections averaged over all held-out views of each scene and then over the five organs. PSNR2D is $20 \log_{10}(1/\sqrt{\text{MSE}})$; SSIM2D uses an 11×11 Gaussian window ($\sigma = 1.5$, $C_1 = 0.01^2$, $C_2 = 0.03^2$).

Implementation details. SPS uses $\alpha = 0.2$,

$\gamma = 1.0$, and 50K initial Gaussians. GAP uses $K = 5$ neighbors, threshold $\tau = 0.015$, maximum pruning ratio $\beta_{\text{prune}} = 0.02$, and gradient threshold $\delta = 0.0002$. ADM uses K-Planes at resolution 64×32 and activates after 15K iterations. All models are trained for 30K iterations.

4.2. Main Results

Table 1 reports PSNR2D across all five organs and three sparsity levels. **XRA-GS** achieves the best five-organ average at every sparsity level. Gains over R2-Gaussian are $+0.11/ +0.39/ +0.07$ dB under 2/3/4-view settings; gains over X-Field* are $+0.87/ +3.25/ +2.79$ dB. The most pronounced margin occurs at 3 views, consistent with our central argument: capacity misallocation is most directly addressable when projection evidence is sufficient to guide redistribution yet too limited for standard densification to converge to a spatially consistent solution.

Table 2 reports SSIM2D averages for the three methods with accessible logs under our unified protocol. **XRA-GS** leads at 2- and 3-view settings and ties R2-Gaussian at the third decimal place in 4-view, confirming the same trend as PSNR2D.

Figure 6 provides qualitative comparisons across all five organs under the 2-view setting. The natural-scene sparse-view baselines (CoR-GS, DNGaussian, FSGS) produce characteristic boundary leakage and streak artifacts under tomographic projection; the direct X-ray baselines (X-Gaussian, R2-Gaussian, X-Field*) preserve global attenuation structure more faithfully. **XRA-GS** further reduces background diffusion and local over-accumulation near high-contrast anatomical boundaries, consistent with the quantitative gains in Table 1.

Figure 7 provides two complementary analyses. The upper portion enlarges the Chest and Foot regions marked in Figure 6, isolating boundary and

Table 1: Main PSNR2D comparison under the unified five-organ, 2/3/4-view protocol. Blank entries denote same-protocol results that are unavailable in the source logs, so we do not estimate or extrapolate them. Best and second-best results are highlighted in bold and underline, respectively.

Method	Chest			Head			Abdomen		
	2v	3v	4v	2v	3v	4v	2v	3v	4v
DNGaussian	14.30	16.02	15.76	15.72	18.04	16.24	19.33	23.51	24.72
CoR-GS	14.96	18.55	20.29	16.24	20.36	23.40	18.50	22.45	23.03
FSGS	17.86	19.99	21.03	19.43	21.53	24.91	18.75	23.34	23.63
X-Gaussian	17.79	20.29	20.90	19.62	21.52	24.46	18.93	23.54	23.85
R2-Gaussian	<u>21.05</u>	<u>26.22</u>	<u>25.47</u>	<u>23.54</u>	<u>26.60</u>	28.66	<u>24.85</u>	<u>29.26</u>	<u>30.56</u>
X-Field*									
XRA-GS	21.26	26.81	25.83	23.63	26.63	<u>28.52</u>	24.97	29.73	30.89

Method	Foot			Pancreas			Avg		
	2v	3v	4v	2v	3v	4v	2v	3v	4v
DNGaussian	20.17	23.51	19.33	16.15	22.43	25.47	17.13	20.70	20.30
CoR-GS	20.17	25.53	26.83	16.15	23.25	23.96	17.20	22.03	23.50
FSGS	<u>22.17</u>	25.37	27.08	<u>23.03</u>	25.27	26.22	20.25	23.10	24.57
X-Gaussian	22.34	25.48	26.92	23.21	25.12	25.55	20.38	23.19	24.34
R2-Gaussian	19.39	<u>28.48</u>	30.04	17.83	<u>28.61</u>	30.90	<u>21.33</u>	<u>27.83</u>	<u>29.13</u>
X-Field*							20.57	24.97	26.41
XRA-GS	19.55	28.49	<u>29.86</u>	17.80	29.43	30.90	21.44	28.22	29.20

structure recovery at the local scale. The lower portion evaluates a Chest 2-view case at three novel angles (0° , 45° , 90°), confirming that the advantage of **XRA-GS** over R2-Gaussian is stable across viewing directions rather than concentrated at a single angle.

Same-protocol comparison with X-Field. The X-Field* row in Table 1 confirms that the gains of **XRA-GS** are not an artifact of protocol asymmetry. Re-evaluated under our identical 2/3/4-view set-

ting, X-Field* trails **XRA-GS** by a consistent average margin across all sparsity levels.

4.3. Module and Initialization Analysis

Table 3 reports a progressive ablation starting from the R2-Gaussian backbone. SPS raises the 2-view average PSNR2D from 21.27 to 21.44 dB, establishing that path-anchored initialization provides a meaningful head start in the most under-determined

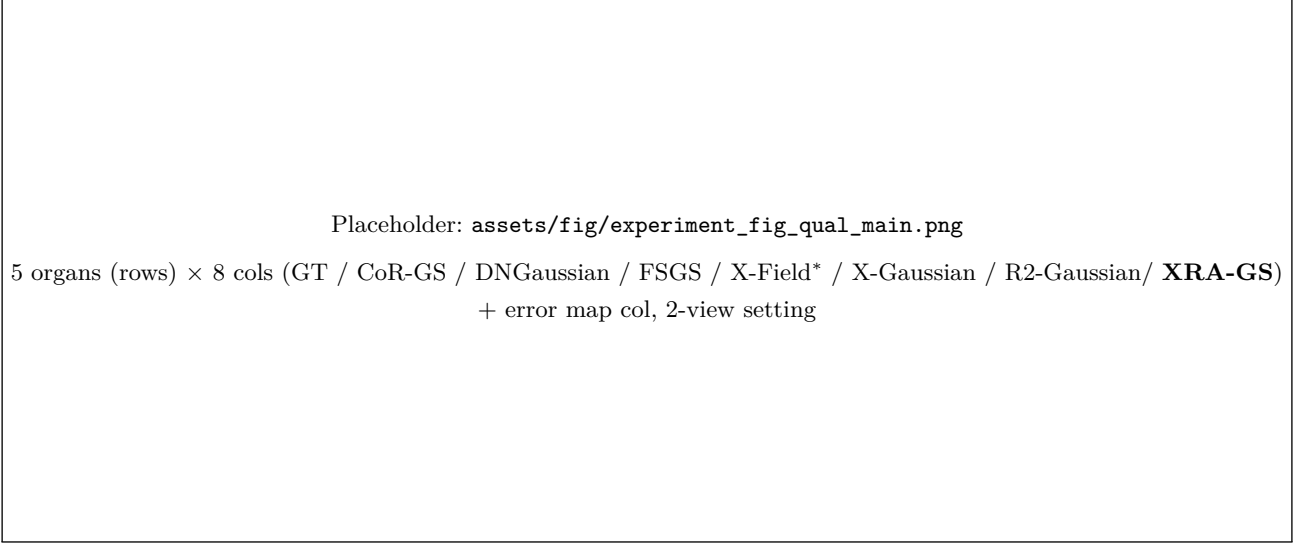


Figure 6: Qualitative comparison on five organs, 2-view setting. Rows: Chest, Head, Abdomen, Foot, Pancreas (top to bottom). Columns: GT, CoR-GS, DNGaussian, FSGS, X-Field*, X-Gaussian, R2-Gaussian, and **XRA-GS** (Ours, last column). Natural-scene references produce streak artifacts and boundary leakage; direct X-ray baselines preserve global attenuation structure; **XRA-GS** further suppresses local over-accumulation and background diffusion. Rightmost column: error maps (inferno, shared vmax). Red boxes mark regions enlarged in Figure 7.

Table 2: SSIM2D comparison (five-organ average). Per-organ SSIM2D logs for CoR-GS, DNGaussian, FSGS, and X-Gaussian are unavailable under our unified 2/3/4-view protocol; we therefore report same-protocol averages only for the three methods with accessible logs. **Bold:** best; underline: second-best.

Method	2v Avg.	3v Avg.	4v Avg.
X-Field*	0.717	0.850	0.880
R2-Gaussian [ZLC*24]	<u>0.794</u>	<u>0.903</u>	0.924
XRA-GS (Ours)	0.797	0.904	0.924

regime. GAP delivers the largest PSNR2D increment, improving the 3-view average from 28.01 to 28.22 dB by reclaiming capacity from over-densified boundary regions. The final **XRA-GS** row matches or improves the available PSNR2D measurements

Table 3: Progressive component ablation (five-organ average). Modules are added incrementally on top of the R2-Gaussian backbone. GAP delivers the largest available PSNR2D increment, while SPS benefits the most under-determined 2-view regime. Intermediate SSIM2D values are pending from experiment logs, so the current table supports PSNR2D trends only for intermediate configurations. **Bold:** best per column.

Configuration	PSNR2D [†] (dB)			SSIM2D [†]		
	2v	3v	4v	2v	3v	4v
Baseline (R2-Gaussian)	21.27	27.80	29.10	–	–	–
+ SPS	21.44	28.01	29.16	–	–	–
+ SPS + GAP	21.44	28.22	29.20	–	–	–
XRA-GS (SPS + GAP + ADM)	21.44	28.22	29.20	0.797	0.904	0.924

SSIM2D for intermediate configurations is pending from experiment logs.

across all view counts, while intermediate SSIM2D values are pending from the experiment logs.

Figure 8 translates the progressive ablation into

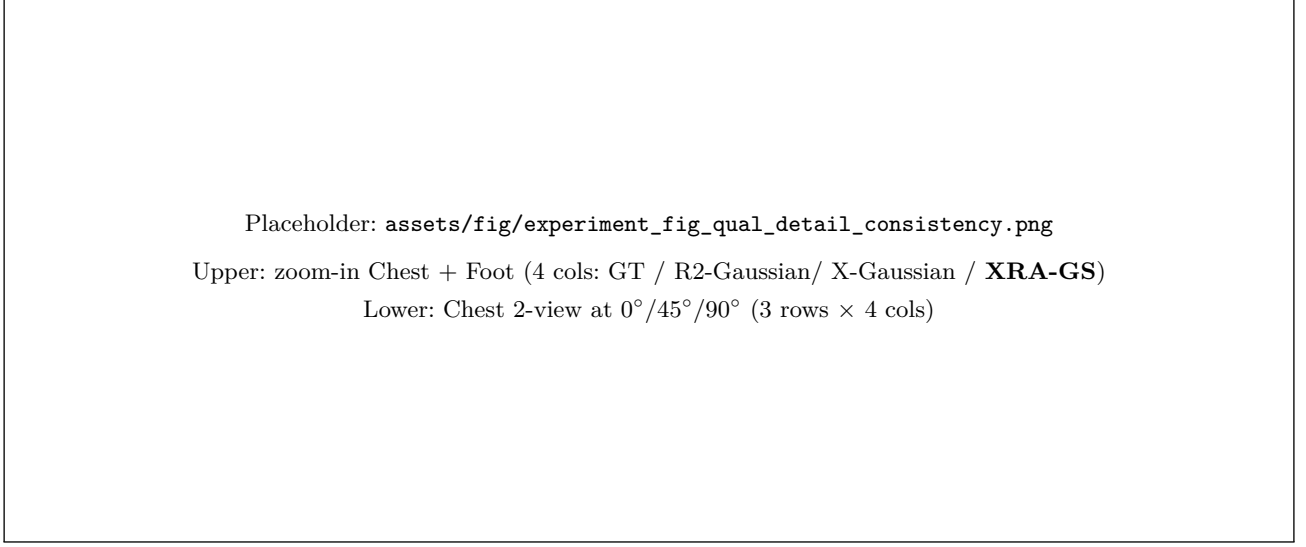


Figure 7: Local detail and cross-view consistency. Upper half: Zoom-ins from the red boxes in Figure 6 for Chest (left) and Foot (right), comparing GT, R2-Gaussian, X-Gaussian, and **XRA-GS**. **XRA-GS** reduces streak spread and boundary diffusion in both regions. Lower half: A Chest 2-view case rendered at 0° , 45° , and 90° . The quality advantage of **XRA-GS** persists across all three angles, indicating consistent global attenuation reconstruction rather than angle-specific improvement.

local visual evidence. Residual maps $|\hat{I} - I_{\text{gt}}|$ are shown below each prediction. This visual analysis is used to inspect where the progressive modules change the reconstruction, while the corresponding intermediate SSIM2D values remain pending from the experiment logs.

Figure 9 is reserved for visualizing the cumulative effect of the three modules on Gaussian spatial distribution once the corresponding Gaussian-center and count logs are available. Until those logs are confirmed, the mechanism is evaluated through the available quantitative ablation in Table 3.

Table 4 (left) isolates the initialization strategy. SPS-guided seeding outperforms random initialization at all sparsity levels; the full **XRA-GS** further improves upon SPS-only through GAP and ADM, confirming that path-anchored initialization and capacity recovery are complementary operations.

4.4. View-count Analysis

Table 4 (right) reports the view-count trend for four representative methods. **XRA-GS** achieves the highest five-organ average PSNR2D at every sparsity level. The relative gain over R2-Gaussian is largest at 3 views, where capacity misallocation is most directly observable; at 4 views the reconstruction-consistency regularization in R2-Gaussian partially closes the gap, consistent with the applicability boundary discussed in Section 4.7.

4.5. Computational Cost

Table 5 is reserved for training and inference cost under the 3-view setting. The current manuscript keeps the table structure in place, but Time, final Gaussian count, GPU memory, and FPS must be filled from experiment logs before making quantitative efficiency claims.

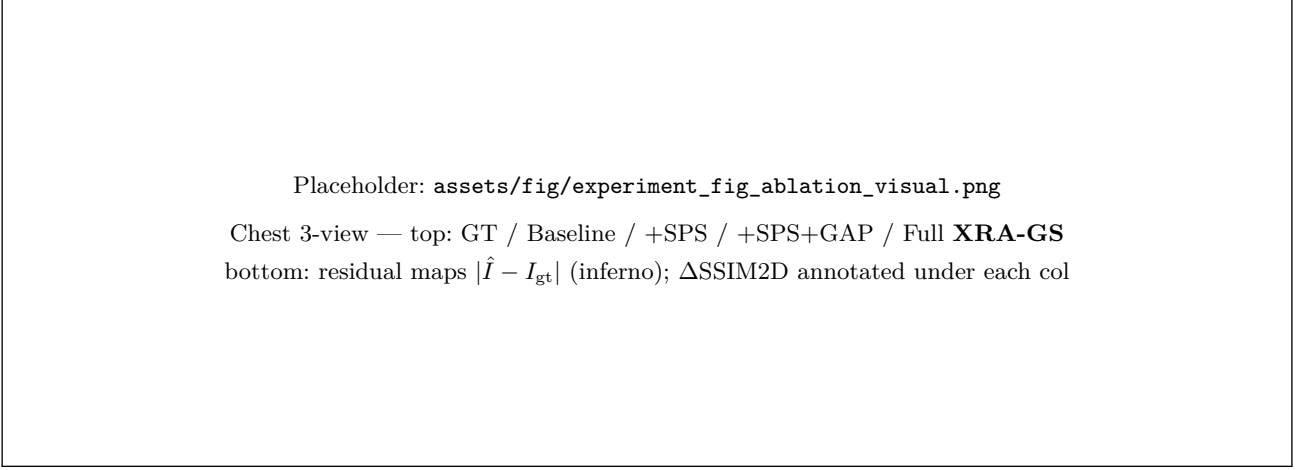


Figure 8: Progressive module ablation, Chest 3-view. Top row: rendered projections for GT, Baseline (R2-Gaussian), +SPS, +SPS+GAP, and full **XRA-GS**. Bottom row: residual maps $|\hat{I} - I_{gt}|$ (inferno, shared vmap with Figure 6). Quantitative intermediate SSIM2D values are pending from experiment logs and are therefore not used for module-level claims here.

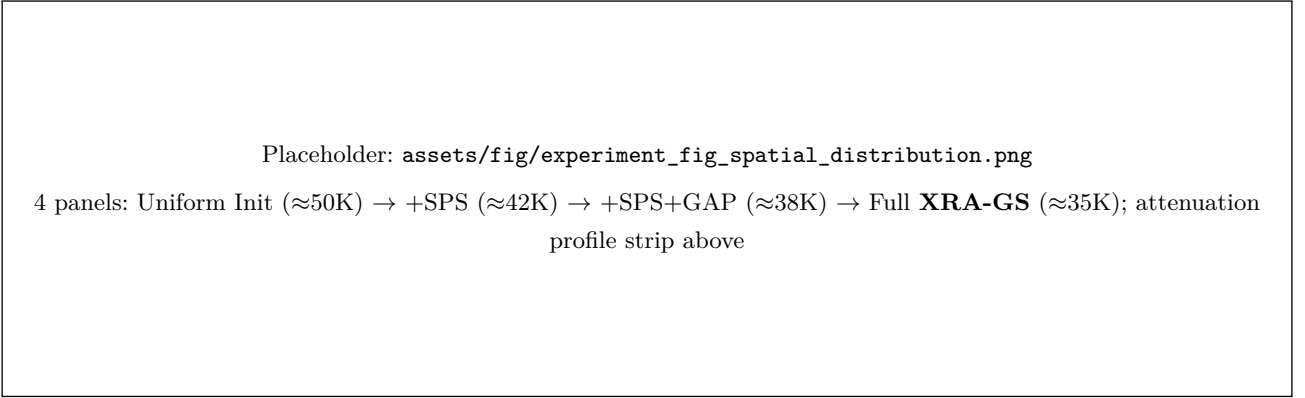


Figure 9: Gaussian spatial distribution evolution. Four panels (left to right): uniform initialization, after SPS, after GAP pruning, and full **XRA-GS**, for a representative Chest organ. Top strip: FDK attenuation profile (magma shading). The final figure will use logged Gaussian centers and counts rather than schematic estimates.

4.6. Hyperparameter Sensitivity

be assessed once the sweep metrics are available from the experiment logs.

Figure 10 is reserved for local sweeps of four key hyperparameters in the Chest 3-view setting: the SPS mixture coefficient α , the GAP proximity threshold τ , the GAP pruning ratio β_{prune} , and the ADM activation iteration. The default configuration ($\alpha = 0.2$, $\tau = 0.015$, $\beta_{\text{prune}} = 2\%$, ADM activation at 15K) will

Table 4: Left: Initialization strategy ablation (five-organ average PSNR2D, dB). SPS-guided seeding outperforms random initialization at all sparsity levels; the full **XRA-GS** further gains from GAP and ADM, confirming that path-anchored initialization and capacity recovery are complementary. **Right: View-count trend** for four representative methods (five-organ average PSNR2D, dB). **XRA-GS** leads at every sparsity level; the relative advantage over R2-Gaussian is largest at 3 views, where capacity misallocation is most directly observable. **Bold:** best; underline: second-best.

Initialization	2v	3v	4v
Random (no SPS)	21.27	27.80	29.10
SPS-guided (no GAP, ADM)	<u>21.44</u>	<u>28.01</u>	<u>29.16</u>
XRA-GS (full)	21.44	28.22	29.20

Five-organ avg PSNR2D $\uparrow$ (dB). Per-organ and SSIM2D data in supplementary.

Method	2v	3v	4v
FSGS [ZFJW24]	20.25	23.10	24.57
X-Gaussian [CLW*24]	20.38	23.19	24.34
R2-Gaussian [ZLC*24]	<u>21.33</u>	<u>27.83</u>	<u>29.13</u>
XRA-GS (Ours)	21.44	28.22	29.20

Five-organ avg PSNR2D $\uparrow$ (dB). CoR-GS and DNGaussian omitted for brevity

(full data in Table 1).

Table 5: Computational cost comparison under the 3-view setting (five-organ average). *Train Time*: end-to-end training minutes. *#Gaussians*: final primitive count ($\times 10^3$) after densification/pruning. *GPU Mem*: peak memory during training (GB). *FPS*: rendering frames per second at inference. *PSNR2D*: five-organ average (dB), from Table 1. *Cost columns* will be filled from matched experiment logs before quantitative efficiency claims are made. **Bold**: best per column.

Method	Time $\downarrow$ (min)	#Gauss. $\downarrow$ (K)	Mem. $\downarrow$ (GB)	FPS $\uparrow$	PSNR2D $\uparrow$ (dB)
X-Gaussian [CLW*24]	—	—	—	—	23.19
R2-Gaussian [ZLC*24]	—	—	—	—	27.83
XRA-GS (Ours)	—	—	—	—	28.22

“—” indicates values pending from experiment logs requested in

`assets/ask/experiment_2026-05-23_T01_T02_T04_main_metrics`

`_efficiency.md`. PSNR2D is taken from Table 1.

Placeholder: `assets/fig/experiment_fig_hparam.png`

2 $\times$ 2 panels: (a) SPS α ; (b) GAP τ ; (c) GAP

β_{prune} ; (d) ADM activation

Primary y-axis: SSIM2D (solid Crimson); secondary:

PSNR2D (dashed); $\bullet$ = default

Figure 10: Hyperparameter sensitivity, Chest 3-view. Each panel sweeps one hyperparameter with all others fixed at the default (marked $\bullet$). Solid Crimson: SSIM2D $\uparrow$; dashed: PSNR2D $\uparrow$. (a) SPS mixture coeff. $\alpha \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$; (b) GAP threshold τ ; (c) GAP pruning ratio $\beta_{\text{prune}} \in \{0, 1, 2, 5, 10\}\%$; (d) ADM activation iteration (fraction of 30K iters). The final sweep figure will be populated from experiment logs.

4.7. Discussion and Limitations

The experiments confirm that in the moderately sparse regime, the primary bottleneck is capacity allocation rather than Gaussian count. Gains are most pronounced at 3 views, where sufficient projection evidence exists for path-anchored initialization and boundary de-redundancy to take effect, yet too little evidence for standard densification to converge to a spatially consistent solution.

Two boundaries constrain the current method. At 2 views the FDK reconstruction used by SPS is insufficiently reliable, so the initialization advantage partially erodes. At 4 views the reconstruction-consistency regularization in R2-Gaussian is strong enough to compensate for boundary-chasing effects, reducing the relative advantage of **XRA-GS**. Figure 11 illustrates both failure modes: in Foot 2-view and Pancreas 2-view, fine-grained high-frequency detail remains unrecoverable when projection evidence

is fundamentally insufficient, regardless of initialization or capacity management.

Several directions remain open. First, SPS relies on FDK quality, which degrades sharply below two views; learned or uncertainty-aware initializers could relax this constraint. Second, the GAP threshold and pruning ratio are fixed organ-agnostically; organ-adaptive scheduling may improve per-organ performance. Third, all experiments assume parallel-beam geometry; extension to cone-beam and dynamic CT acquisition is left for future work. Additional per-organ SSIM2D breakdowns and single-module ablations will be included only after the corresponding same-protocol logs are confirmed.

5. Conclusion

XRA-GS demonstrates that visible-light densification rules impose a systematic capacity-allocation failure in transmissive X-ray imaging that persists even after the rendering equation is physically corrected. By intervening at three stages of Gaussian evolution—path-anchored initialization (SPS), boundary redundancy recovery (GAP), and position-dependent local refinement (ADM)—the framework redirects representation capacity from over-densified anatomical boundaries toward the interior attenuation-relevant structures that conventional gradient-driven growth neglects. Across a five-organ benchmark with 2/3/4 input views against six representative baselines, **XRA-GS** achieves the best average PSNR2D at every sparsity level, with SSIM2D reported for methods whose same-protocol logs are available. The strongest PSNR2D improvement appears in the moderately sparse 3-view regime, where sufficient projection evidence allows capacity reallocation to take effect.

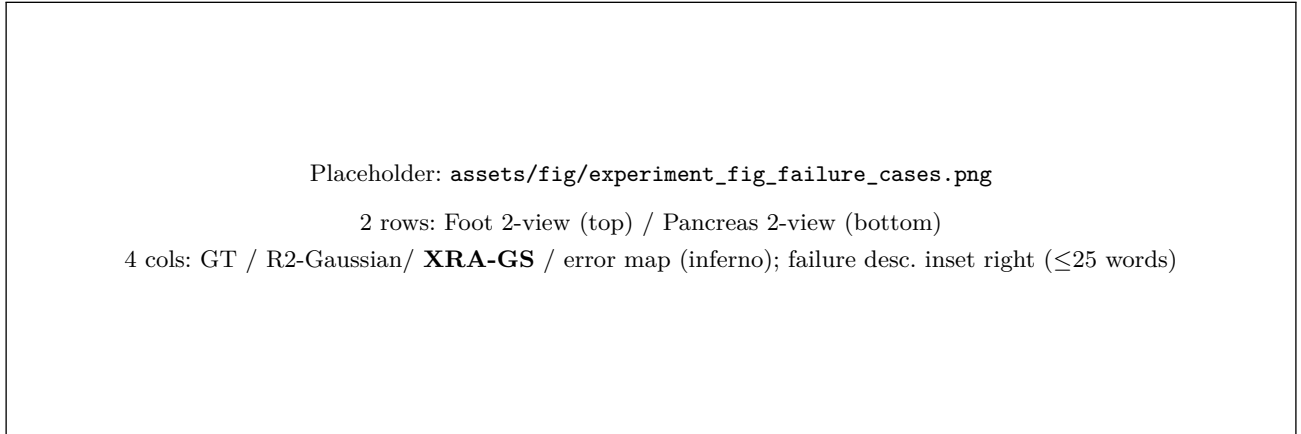


Figure 11: Representative failure cases, 2-view setting. Rows: Foot (top), Pancreas (bottom). Columns: GT, R2-Gaussian, **XRA-GS**, and **XRA-GS** error map (inferno, shared vmax with Figure 6). Zoom-in boxes mark the primary failure region. **XRA-GS** reduces overall error spread relative to R2-Gaussian but cannot resolve fine structural detail when projection evidence is fundamentally absent. Failure descriptions (≤ 25 words) are inset at the right of each row.

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